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WHU-Hi+: A Large-Scale Benchmark Dataset for Fine-Grained Hyperspectral Cross-Image Classification

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***Abstract*—High-spatial–high-spectral-resolution UAV imagery provides detailed spectral–spatial information for fine-grained agricultural mapping, while existing hyperspectral benchmarks remain limited in spatial coverage, category complexity, and evaluation protocols. To address these limitations, we introduce WHU-Hi+, a large-scale UAV-based hyperspectral benchmark for fine-grained hyperspectral image classification. WHU-Hi+ represents a complex agricultural scene and comprises 105 non-overlapping image tiles with 11 crop categories. The dataset is characterized by a fine-grained crop taxonomy, class imbalance, inter-class spectral similarity and intra-class spectral variability, and heterogeneous crop distributions across image tiles. To support comprehensive evaluation, we establish three train–test partitioning protocols including random pixel sampling (Protocol A), within-tile label partitioning (Protocol B), and tile-disjoint partitioning (Protocol C). Protocol B uses entire tiles as inputs but restricts training supervision to one half of each tile, whereas Protocol C further restricts training inputs to designated training tiles, excluding test tiles entirely. Ten representative deep learning methods covering convolutional, Transformer-based, Mamba-based, hyperspectral-specific, and foundation-model architectures are systematically benchmarked. Most methods show high performance under conventional random sampling but substantial degradation under spatially separated settings, showing that benchmark performance depends on the adopted evaluation protocol. WHU-Hi+ provides a challenging benchmark for evaluating fine-grained hyperspectral image classification under diverse spatial partitioning settings. The dataset, protocols, and benchmark code are publicly available at the project website(https://jiangyuelin0615.github.io/WHU-HiPlus/).**

***Index Terms*—hyperspectral remote sensing; UAV hyperspectral imagery; benchmark dataset; fine-grained classification; spatial partitioning.**

I. Introduction

HYPERSPECTRAL remote sensing acquires densely sampled narrow-band spectral information of ground objects, enabling fine-grained land-cover classification through enhanced material discrimination capability compared with conventional multispectral imagery [1]–[7]. Advances in unmanned aerial vehicle (UAV)-based hyperspectral imaging systems have enabled the acquisition of high-spatial–high-spectral-resolution (H^2) imagery, supporting applications in precision agriculture and vegetation monitoring [8]–[13]. Although recent deep learning models have further improved hyperspectral image classification performance [7], [14]–[17], their reported performance depends on the quality and evaluation protocols of benchmark datasets. Datasets that represent spectral–spatial complexity, spatial heterogeneity, and category diversity are therefore essential for reliable algorithmic comparison and comprehensive performance evaluation [1], [18].

Recent advances in UAV-based hyperspectral sensing have facilitated the development of H^2 benchmark datasets with higher spatial resolution, greater scene diversity, and finer semantic granularity, as summarized in Table I. Existing datasets cover various applications, including mining and urban environments [19]–[21], ecological regions [22], agricultural monitoring [23]–[26], and forest mapping [27]. Among them, WHU-Hi and UC-HSI provide detailed crop taxonomies and annotations for fine-grained agricultural classification, while Luojia-HSSR [28] covers a larger area with diverse land-cover categories.

Existing H^2 hyperspectral benchmarks exhibit two major limitations. First, achieving both broad spatial coverage and fine-grained crop categorization remains challenging. Large-scale datasets such as Luojia-HSSR encompass diverse land-cover categories, while fine-grained agricultural datasets such as WHU-Hi and QUH typically cover localized regions. Limited spatial coverage restricts the representation of heterogeneous crop distributions and intra-class spectral variability across agricultural areas [29], [30].

Second, many hyperspectral benchmarks employ within-image random pixel sampling, where training and test samples are spatially intermingled or closely located in homogeneous land-cover regions. Previous studies have shown that such spatial dependence may lead to optimistic performance estimates [31], [32]. Consequently, evaluation results obtained under random sampling may not fully characterize model performance across different spatial configurations. This motivates the introduction of complementary spatially disjoint partitioning protocols to provide a more comprehensive evaluation of hyperspectral classification methods.

To address these limitations, we construct WHU-Hi+, a large-scale UAV-based hyperspectral benchmark for fine-grained image classification. Unlike conventional hyperspectral benchmarks that are typically organized as a single image, WHU-Hi+ is organized into 105 non-overlapping image tiles. This multi-tile organization enables the construction of different train–test spatial partitioning protocols for cross-image

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TABLE I
EXISTING DATASETS WITH HIGH SPATIAL AND SPECTRAL RESOLUTION

Dataset	Platform	Spectral Resolution	Spatial Resolution	Bands	Image Size	Category	Sensor	Release Year
Xuzhou HYPSEX	Aerial	8nm	0.73m	436	500×260	9	HYSPEX VNIR-1600 / HYSPEX SWIR-384	2019
AeroRIT	Aerial	-	0.4m	372	1973×3975	5	Headwall Micro E	2020
State-Owned Forest Farm	Aerial	3.3nm	1m	125	572×906	9	LiCHy	2020
Matiwan Village	Aerial	-	0.5m	250	3750×1580	19	AMMHS	2020
Luojia-HSSR	Aerial	$\leq$5nm	0.75m	249	161 km^2	23	AMMHS	2022
SDFC	Aerial	10nm	0.5m	63	1201×601	20	HAHS	2023
WHU-Hi-HanChuan	UAV	6nm	0.109m	274	1217×303	16	Headwall Nano-HyperSpec	2020
WHU-Hi-HongHu	UAV	6nm	0.043m	270	940×475	22	Headwall Nano-HyperSpec	2020
WHU-Hi-LongKou	UAV	6nm	0.463m	270	550×400	9	Headwall Nano-HyperSpec	2020
UAV-HSI-Crop-A	UAV	3.3nm	0.1m	200	2332×959	29	Resonon Pika L	2022
UAV-HSI-Crop-B	UAV	3.3nm	0.1m	200	864×1618	29	Resonon Pika L	2022
QUH-Tangdaowan	UAV	3.5nm	0.15m	176	1740×860	18	Gaiasky mini2-VN	2023
QUH-Qingyun	UAV	3.5nm	0.15m	176	880×1360	6	Gaiasky mini2-VN	2023
QUH-Pingan	UAV	3.5nm	0.1m	176	1230×1000	10	Gaiasky mini2-VN	2023
UC-HSI-Crop Type	UAV	2.1nm	0.011m	300	100×100 (533+353)	10	Pika-L HSI	2024
UC-HSI-Crop Growth Stage	UAV	2.1nm	0.011m	300	200×150 (366+308)	5	Pika-L HSI	2024
WHU-Hi+	UAV	6nm	0.135m	270	256×256 (105)	11	Headwall Nano-HyperSpec	2026

classification. The dataset further presents several challenges for fine-grained crop classification, including spectral similarity among crop categories, intra-class spectral variability, class imbalance, and heterogeneous crop distributions across image tiles. The main contributions of this work are summarized as follows.

1) A large-scale fine-grained H^2 agricultural hyperspectral dataset organized into 105 non-overlapping image tiles, characterized by spectrally similar crop categories, severe class imbalance, substantial intra-class spectral variability, and heterogeneous crop distributions.
2) A multi-protocol evaluation framework for hyperspectral cross-image classification, comprising a conventional random-sampling reference, a within-tile spatially disjoint setting, and a tile-disjoint cross-image setting, for systematically assessing hyperspectral classification performance under different spatial arrangements.
3) Comprehensive benchmarking of ten representative deep learning methods covering convolutional, Transformer-based, Mamba-based, hyperspectral-specific, and foundation-model architectures, providing reference results and comparative analysis under different train–test spatial partitioning protocols.

The remainder of this paper is organized as follows. Section II introduces the acquisition, processing procedures, and characteristics of the WHU-Hi+ dataset. Section III introduces three evaluation protocols. Section IV presents the experimental settings, evaluation metrics, and results. Section V provides further discussion, and Section VI concludes the paper.

II. THE WHU-HI+ DATASET

A. Study Area

The WHU-Hi+ study area is located north of Zouzhuang Village, Guanggu District, Wuhan, Hubei Province, China. The region features flat terrain, fertile soils, and fragmented agricultural parcels cultivated by smallholder farmers. Variations in crop types, planting schedules, and farming practices result in heterogeneous crop distributions across the study area. In particular, differences in sowing dates lead to variations in phenological stages even within the same crop species, contributing to substantial intra-class spectral–spatial variability. Together, these characteristics create a challenging setting for fine-grained crop classification.

B. Data Acquisition and Preprocessing

The WHU-Hi+ dataset was acquired on January 12, 2021, using a Headwall Nano-Hyperspec sensor mounted on a DJI Matrice 600 Pro UAV platform over the study area. Ground truth annotations were collected through field surveys using GPS devices to record crop distributions within the study region.

The raw hyperspectral data were processed through radiometric calibration, reflectance correction, geometric correction, and mosaicking. Radiometric calibration was conducted in HyperSpec software to convert raw digital number (DN) values into radiance measurements according to the sensor calibration model:

$$\mathrm{DN} = (L_1 G_1 + L_2 G_2) \times ET + DF \tag{1}$$

where L_1 and L_2 denote first- and second-order radiance components, G_1 and G_2 are the corresponding system gains, ET is the exposure time, and DF represents dark-field measurements.

Surface reflectance was subsequently estimated using empirical line calibration with three reference panels of known reflectance values deployed within the scene. Geometric correction was performed based on the collinearity equations using GPS/IMU measurements and ground control points (GCPs).

After preprocessing, individual flight strips were co-registered and mosaicked using a high-resolution reference image. The final mosaic was uniformly cropped into non-overlapping image tiles, and tiles containing less than 15% labeled pixels were discarded. This procedure resulted in 105 image tiles for subsequent benchmark construction.

C. Data Characteristics

1) Category System and Sample Distribution: The study area of the WHU-Hi+ dataset represents a complex agricultural scene, as illustrated in Fig. 1, encompassing a total of 11 crop categories. These include a variety of common crops such as rape, lettuce, broad bean, pea, nori moss, and cauliflower, as

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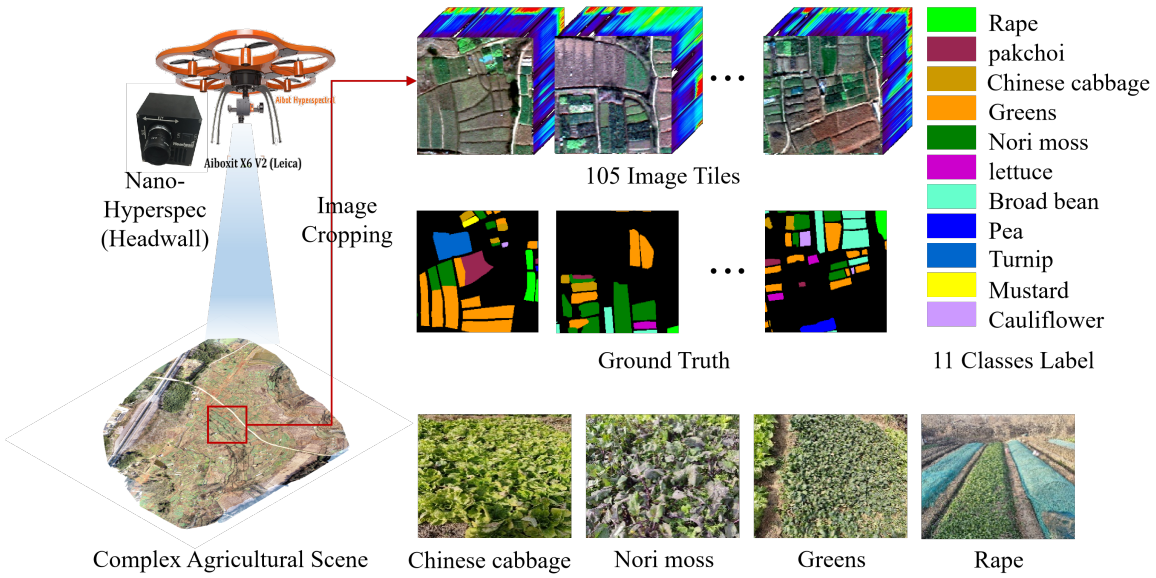


Fig. 1. WHU-Hi+ Dataset.

TABLE II
CLASS ANNOTATION STATISTICS OF WHU-HI+ DATASET.

No.	Class	Pixel Count	Proportion (%)	No.	Class	Pixel Count	Proportion (%)
C1	Rape	888999	53.37	C7	Broad bean	94606	5.68
C2	Pakchoi	38016	2.28	C8	Pea	67107	4.03
C3	Chinese cabbage	100132	6.01	C9	Turnip	35832	2.15
C4	Greens	151909	9.12	C10	Mustard	27611	1.66
C5	Nori moss	181335	10.89	C11	Cauliflower	36131	2.17
C6	Lettuce	44031	2.64				

well as spectrally similar crop pairs such as pak choi and Chinese cabbage, forming a fine-grained class taxonomy. From the perspective of sample quantity, the class distribution is highly imbalanced. Rape alone accounts for more than half of all labeled pixels, indicating pronounced class imbalance. Detailed per-class sample statistics are provided in Table II.

2) Inter-Class Spectral Similarity and Intra-Class Spectral Variability: The mean spectral profiles of all crop classes in WHU-Hi+ are shown in Fig. 2(a). Since all categories correspond to crop types, substantial spectral overlap exists among different classes, particularly between spectrally similar categories, posing challenges for fine-grained discrimination.

In addition, WHU-Hi+ exhibits pronounced intra-class spectral variability. The multi-strip UAV acquisition covers spatially heterogeneous agricultural regions, where differences in illumination, viewing geometry, field conditions, and crop growth status may contribute to variations in spectral responses. Consequently, the same crop category can exhibit different spectral characteristics across image tiles.

To illustrate this variability, Fig. 2(b) shows the per-tile mean spectral curves of rape across the WHU-Hi+ dataset. Although the overall spectral trend remains generally consistent, substantial reflectance fluctuations are observed within the 750–1000 nm range, with substantial differences in reflectance magnitude across image tiles. These variations demonstrate substantial intra-class spectral variability even within the dominant category of WHU-Hi+.

To quantitatively characterize the spectral similarity and class separability among fine-grained crop categories in WHU-Hi+, we computed the Spectral Angle Mapper (SAM) and Jeffries–Matusita (JM) distance for all class pairs. SAM measures the angular difference between class-wise mean spectral signatures, with smaller values indicating greater spectral-shape similarity. The mean spectrum of each class was computed using all its labeled pixels. JM evaluates the statistical separability of class-conditional spectral distributions, ranging from 0 to 2, with higher values indicating greater separability. Both metrics were computed using pre-standardization spectral intensities. For JM estimation, 10,000 labeled pixels were sampled uniformly without replacement from each class. Each class distribution was approximated by a multivariate Gaussian, with Oracle Approximating Shrinkage (OAS) applied to covariance estimation.

Table III presents the five class pairs with the smallest SAM values. Among them, *Pakchoi* and *Cauliflower* exhibit the highest mean spectral-shape similarity, with a SAM of 0.0125 rad and a JM distance of 1.9347. Similarly, *Broad bean* and *Turnip* have a SAM of 0.0132 rad and a JM distance of 1.9001. The small SAM values indicate substantial similarities in the mean spectral shapes of these crop categories, consistent with the closely aligned spectral signatures shown in Fig. 2(a). Meanwhile, their relatively high JM distances indicate considerable statistical separability under the multivariate Gaussian approximation. These descriptive results indicate that similar class-mean spectral shapes can coexist with high estimated distributional separability under the Gaussian approximation;

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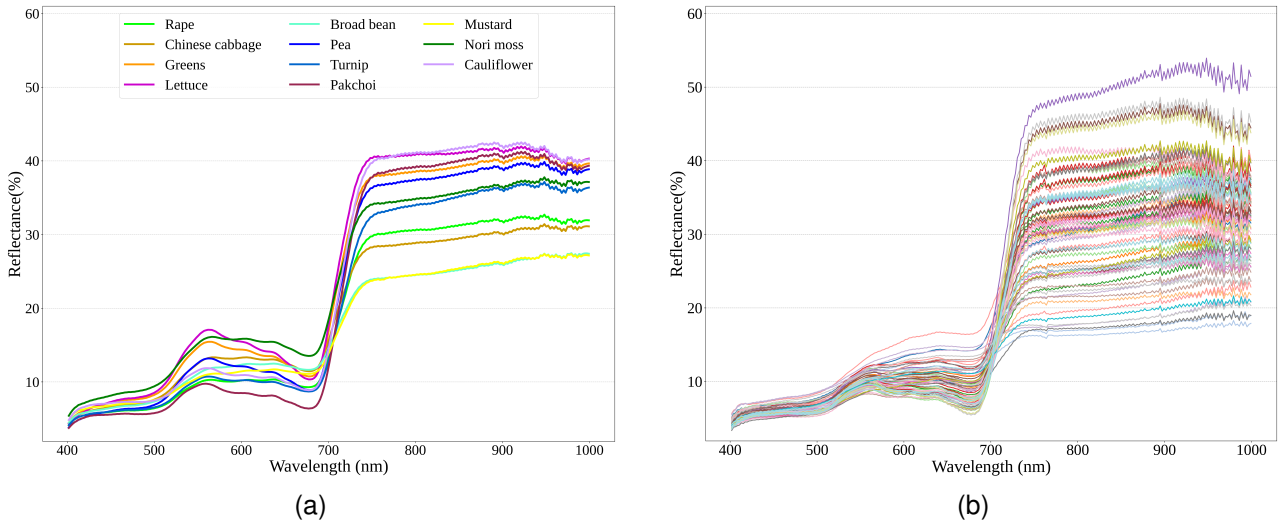


Fig. 2. Spectral characteristics of the WHU-Hi+ dataset. (a) Mean spectral curves of all 11 crop classes. (b) Per-tile mean spectral curves of rape, illustrating intra-class spectral variability across image tiles.

TABLE III
SPECTRAL SIMILARITY AND SEPARABILITY OF REPRESENTATIVE CROP-CLASS PAIRS IN WHU-HI+.

Class Pair	SAM (rad)	JM Distance
Pakchoi vs. Cauliflower	0.0125	1.9347
Broad bean vs. Turnip	0.0132	1.9001
Lettuce vs. Pea	0.0152	1.8649
Rape vs. Nori moss	0.0206	1.9764
Chinese cabbage vs. Mustard	0.0220	1.9526

they do not directly measure classification difficulty under spatial partitioning.

3) Complex Crop Distribution Across Different Images: Most existing H^2 datasets for fine-grained crop classification are constructed from a single image or flight strip, where crop distributions remain spatially fixed. Under such settings, models may partially rely on spatial arrangement patterns rather than learning transferable spectral–spatial representations.

WHU-Hi+ is partitioned into 105 non-overlapping image tiles covering heterogeneous crop distributions across the study area. As illustrated in Fig. 3, both the number of labeled pixels and crop-category composition vary considerably across tiles. Some tiles contain multiple crop categories, whereas others are dominated by a single class. This heterogeneous multi-tile organization provides diverse spatial configurations for constructing cross-image and spatially separated evaluation settings.

4) Data and Code Availability: The WHU-Hi+ dataset is publicly archived in Zenodo under the Creative Commons Attribution 4.0 International (CC BY 4.0) license with the persistent DOI https://doi.org/10.5281/zenodo.23252327. The project website (https://jiangyuelin0615.github.io/WHU-HiPlus/) provides the dataset description, benchmark protocols, download instructions, and a direct link to the archived Zenodo record.

III. DESIGN OF THREE EVALUATION PROTOCOLS

To evaluate hyperspectral crop classification under different train–test configurations, we design three complementary protocols, as illustrated in Fig. 4. Protocol A follows conventional random pixel sampling, whereas Protocols B and C reduce the strong local spatial dependence associated with randomly intermingled training and test samples through within-tile and tile-level spatial partitioning, respectively. Protocol B restricts the regions used for training supervision, while Protocol C further excludes test tiles from training inputs. These protocols provide distinct evaluation settings with different spatial arrangements and sample distributions, enabling systematic assessment of classification performance across different train–test settings.

Protocol A serves as the standard random-sampling baseline. For each image tile, up to N labeled pixels are randomly selected without replacement from each class for training, while all remaining labeled pixels in the same tile are used for testing. Consequently, training and test samples may be spatially intermingled or located in close proximity. This protocol follows the conventional evaluation setting commonly adopted in hyperspectral image classification.

Protocol B adopts a half-tile separation strategy. Each image tile is divided into left and right halves. For each tile, up to N labeled pixels are randomly sampled without replacement from each class in the left half; if fewer than N pixels are available for a class, all available pixels of that class are selected. All labeled pixels in the right half are reserved for testing, resulting in non-overlapping sets of labeled training and test pixels within each tile. Detailed per-class sampling statistics are provided in Table S2 of the Supplementary Material.

Protocol C adopts a tile-disjoint partition based on the spatial arrangement of the 105 image tiles. Let $T_{i,j}$ denote the tile located at row i and column j in the original tile grid. Tiles satisfying $(i+j) \bmod 2 = 0$ constitute the training partition, whereas tiles satisfying $(i+j) \bmod 2 = 1$ constitute

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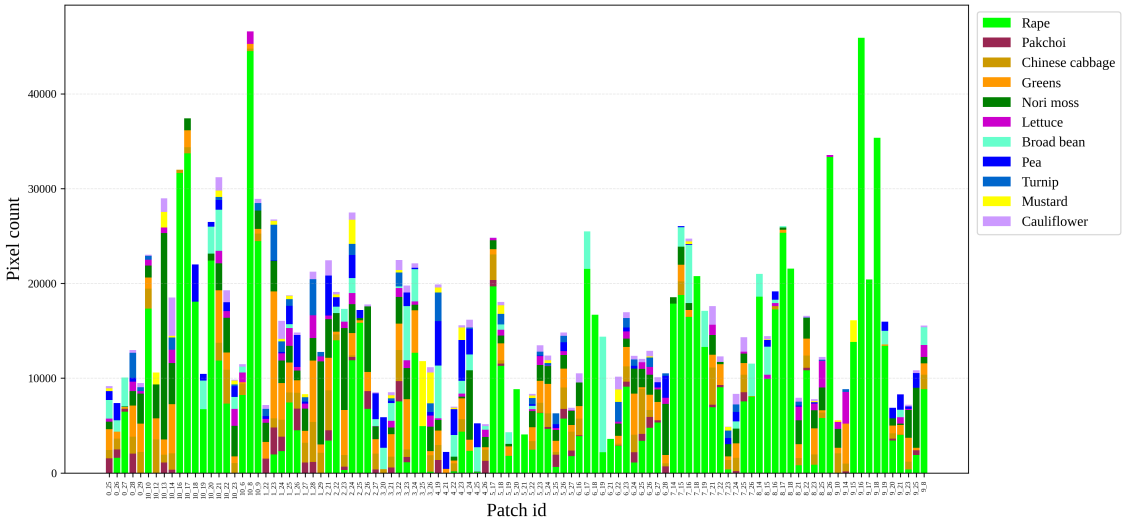


Fig. 3. Per-tile pixel-count distribution across all 11 classes, illustrating heterogeneous class composition and labeled-pixel distributions across image tiles.

the test partition. Training samples are randomly selected only from the training tiles, while all labeled pixels in the test tiles are used for evaluation. No training tile contributes any pixels to the test set, resulting in tile-level non-overlap between the training and test partitions. In contrast to the within-tile separation in Protocol B, Protocol C evaluates cross-image classification with training and test samples drawn from disjoint tiles of the same mosaic.

IV. Experiments

This section conducts benchmark experiments on the WHU-Hi+ dataset using representative deep learning methods for fine-grained crop classification. The evaluated models are compared under different spatial partitioning protocols to examine classification performance under different train–test partitions. The experiments are intended to provide reference results for WHU-Hi+ and to analyze the sensitivity of existing methods to different sampling strategies.

A. Experimental Setup

The experiments are conducted on the WHU-Hi+ dataset, which comprises 11 crop categories and 105 image tiles of 256×256 pixels. For each protocol, class-wise stratified sampling without replacement is used to construct the training set. For each eligible image tile, up to $N \in \{25, 50, 100, 200\}$ labeled pixels are sampled for each class according to the spatial constraints defined by each protocol.

Model performance is evaluated using mean intersection over union (mIoU), mean accuracy (mAcc), and overall accuracy (aAcc). All experiments are repeated with five random seeds (42–46), and results are reported as mean and standard deviation. All methods are implemented within the MMSegmentation framework and evaluated on an NVIDIA TITAN RTX GPU with 24 GB of memory.

B. Comparison Models

To evaluate different model architectures on WHU-Hi+, ten representative deep learning models are selected, covering convolutional networks, Transformer- and Mamba-based architectures, hyperspectral-specific networks, and foundation models.

ConvNeXt [33] is a convolutional architecture developed by modernizing a ResNet design with large kernels, inverted bottlenecks, GELU activations, and layer normalization. We use ConvNeXt-B with a UPerHead for dense prediction.

SegNeXt [34] uses multi-scale convolutional attention to encode spatial context for semantic segmentation. We use the MSCAN-T backbone with a LightHamHead.

SegFormer [35] combines a hierarchical Transformer encoder without explicit positional encoding with a lightweight MLP decoder that aggregates multi-scale features. We use the MiT-B3 backbone with a SegFormerHead.

MambaHSI [36] uses spatial and spectral Mamba modules to model long-range dependencies in hyperspectral imagery. It provides a state-space-model baseline for spectral–spatial classification.

FPGA [37] is a patch-free global learning framework for end-to-end hyperspectral image classification. Here, it is applied to individual image tiles, avoiding sliding-window classification centered on each labeled pixel.

S^{2}AiANet [38] uses spatial–spectral attention in attention to jointly model spatial and spectral information, together with multi-scale attention for contextual feature extraction.

S^{3}ANet [39] models spectral, spatial, and scale information through attention mechanisms for crop classification in H^2 imagery. The present benchmark uses the dense adaptation listed in Table IV and the common pixel-wise cross-entropy training objective described below.

SSFCN [40] uses spectral and spatial fully convolutional branches and fuses their features for dense hyperspectral clas-

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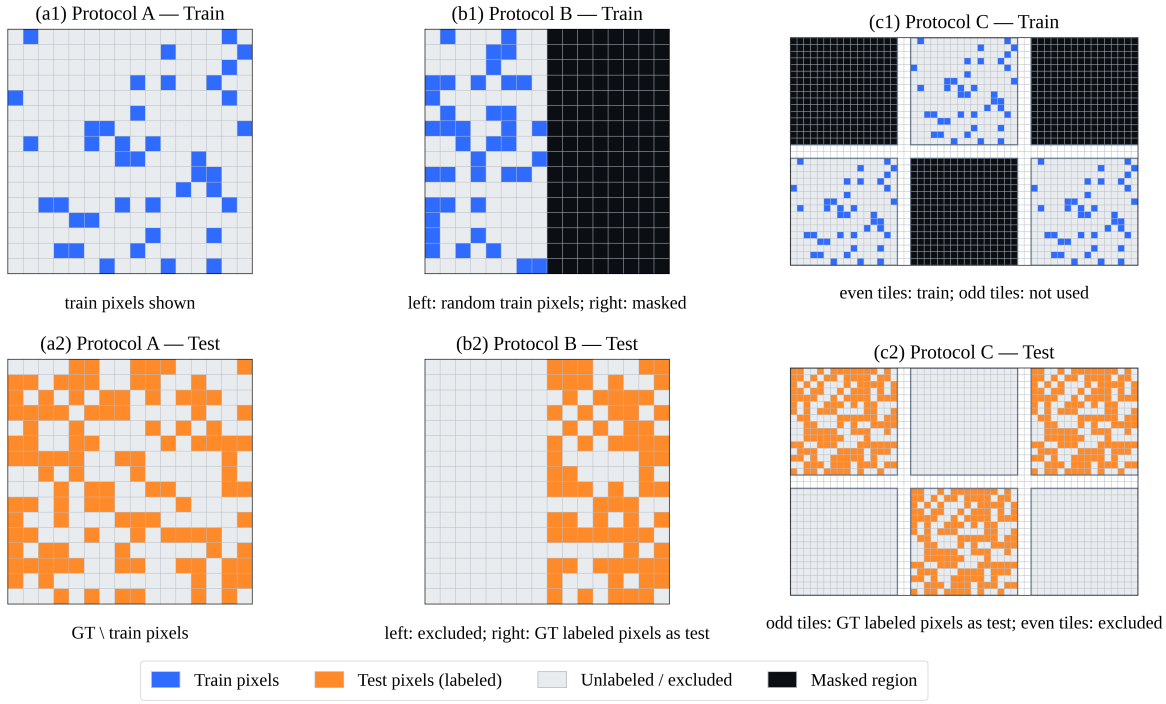


Fig. 4. Schematic illustration of the three protocols: (a) random pixel sampling (Protocol A), (b) half-tile separation (Protocol B), and (c) tile-disjoint partition (Protocol C).

sification. The evaluated adaptation is summarized in Table IV.

HyperSIGMA [41] is a pretrained hyperspectral foundation model that combines spatial and spectral representations. We evaluate a fully fine-tuned adaptation with SpatViT–SpecViT feature fusion and a pixel-wise prediction head.

HyperFree [42] uses a learned wavelength weight dictionary to accommodate different spectral configurations and was developed to support tuning-free applications. In this benchmark, we instead evaluate its pretrained representations using full fine-tuning with a SegFormerHead.

Table IV summarizes the initialization, dense prediction architecture, trainable parameters, and optimization settings of the evaluated models. Full fine-tuning (Full FT) refers to the optimization of all parameters along the active prediction path, and the parameter counts include all trainable components. For pretrained models with standard RGB input stems, the input layers were adapted to accommodate the 270 spectral bands of WHU-Hi+ and initialized from scratch.

All models were trained for 5,000 iterations using 256×256 input patches and pixel-wise cross-entropy loss. Each experiment was repeated with five random seeds (42–46). A linear warm-up of 500 iterations was followed by polynomial learning-rate decay. Weight decay was set to 10^{-4} for SGD-based models, 0.01 for MambaHSI, and 0.05 for HyperSIGMA.

C. Experimental Results and Analysis

The ten comparison methods are evaluated under Protocols A–C with four training sample sizes ($N = 25$, 50, 100, and 200 samples per class per tile). The quantitative results are reported in Table V, where each result is presented as the mean and standard deviation over five runs. Bold indicates the highest mean value in each row.

1) Quantitative Analysis: Under Protocol A, most methods show consistent improvements as the number of training samples increases, with small standard deviations across the five runs. SegFormer and ConvNeXt already achieve mIoU values above 91% with up to 25 labeled pixels per class per tile, while S^{2}AiANet reaches the highest mIoU of 98.46% at $N = 200$. The performance gains gradually diminish as more training samples are introduced, indicating that most methods approach saturation under random pixel sampling. Overall, the high accuracies obtained under Protocol A demonstrate the effectiveness of existing methods under conventional random sampling.

Under Protocol B, the evaluated methods have substantially lower performance than under Protocol A. ConvNeXt achieves the highest mIoU of 35.59% at $N = 200$, while SegFormer and MambaHSI obtain comparable results of 35.53% and 35.06%, respectively. In contrast to Protocol A, increasing the number of training samples provides only limited improvements under this setting. For example, the mIoU of ConvNeXt remains nearly unchanged from 35.42% at $N = 25$ to 35.59% at $N = 200$, while that of SegFormer increases only from 34.82% to 35.53%. These results show limited gains from increasing the sampling cap within the fixed training regions. Additional labels in those regions do not necessarily improve performance on the right-half test regions.

Under Protocol C, the overall performance is reduced compared with Protocol A. SegFormer achieves the best performance across the four sample sizes, with mIoU increasing from 33.67% at $N = 25$ to 34.54% at $N = 200$. Similar

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TABLE IV
IMPLEMENTATION CONFIGURATIONS OF THE EVALUATED MODELS ON WHU-HI+.

Model	Initialization	Dense Adaptation / Decoder	Params	Optimizer	LR	Batch
ConvNeXt	ImageNet-1K + Full FT	ConvNeXt-B + UPerHead	121.36M	SGD	3×10^{-3}	16
SegNeXt	MSCAN-T + Full FT	MSCAN-T + LightHamHead	4.27M	SGD	3×10^{-3}	16
SegFormer	MiT-B3 + Full FT	MiT-B3 + SegFormerHead	45.44M	SGD	3×10^{-3}	16
MambaHSI	From scratch	Spatial–spectral Mamba + pixel-wise head	0.127M	AdamW	6×10^{-4}	8
FPGA	From scratch	FreeNet + FPGA head	2.69M	SGD	3×10^{-3}	16
S^2AiANet	From scratch	FreeNet + S^2AiA head	5.19M	SGD	3×10^{-3}	16
S^3ANet	From scratch	FreeNet + S^3A head	6.06M	SGD	3×10^{-3}	16
SSFCN	From scratch	SSFCN + pixel-wise head	0.118M	SGD	3×10^{-3}	16
HyperSIGMA	Official MAE + Full FT	SpatViT–SpecViT fusion + pixel-wise head	205.94M	AdamW	6×10^{-5}	2
HyperFree	HyperFree-B + Full FT	Multi-level features + SegFormerHead	152.16M	SGD	3×10^{-3}	16

TABLE V
CLASSIFICATION RESULTS (%) UNDER DIFFERENT PROTOCOLS. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS.

Protocol	N	Metric	ConvNeXt	SegNeXt	SegFormer	MambaHSI	FPGA	S^2AiANet	S^3ANet	SSFCN	HyperSIGMA	HyperFree
Protocol A	25	mIoU	91.82 ± 0.27	89.33 ± 0.87	**91.92 ± 0.38**	81.79 ± 0.41	81.58 ± 0.42	88.29 ± 0.34	71.11 ± 1.17	48.59 ± 0.80	88.65 ± 0.59	84.32 ± 0.70
		mAcc	**96.99 ± 0.15**	96.24 ± 0.33	96.98 ± 0.08	91.79 ± 0.24	91.57 ± 0.35	95.42 ± 0.23	85.46 ± 0.86	67.95 ± 0.79	95.62 ± 0.26	94.72 ± 0.19
		aAcc	97.77 ± 0.13	96.80 ± 0.37	**97.80 ± 0.17**	94.16 ± 0.19	94.30 ± 0.14	96.46 ± 0.17	90.47 ± 0.77	80.73 ± 0.56	96.53 ± 0.22	94.10 ± 0.69
	50	mIoU	95.43 ± 0.29	93.65 ± 0.36	**95.58 ± 0.25**	89.53 ± 0.37	91.00 ± 0.24	94.24 ± 0.19	83.50 ± 0.74	53.77 ± 0.71	93.29 ± 0.11	91.00 ± 0.54
		mAcc	**98.43 ± 0.12**	97.87 ± 0.13	98.41 ± 0.10	95.80 ± 0.28	96.29 ± 0.20	97.92 ± 0.07	92.91 ± 0.40	72.74 ± 0.61	97.58 ± 0.08	97.14 ± 0.09
		aAcc	98.72 ± 0.07	98.17 ± 0.13	**98.80 ± 0.08**	96.73 ± 0.09	97.30 ± 0.05	98.30 ± 0.09	95.11 ± 0.20	83.84 ± 0.50	97.99 ± 0.08	97.08 ± 0.30
	100	mIoU	97.16 ± 0.09	95.42 ± 0.23	**97.27 ± 0.07**	93.67 ± 0.15	95.96 ± 0.13	97.07 ± 0.11	90.77 ± 0.59	55.70 ± 0.84	95.78 ± 0.13	94.24 ± 0.20
		mAcc	99.08 ± 0.04	98.55 ± 0.05	**99.09 ± 0.03**	97.66 ± 0.07	98.56 ± 0.07	99.03 ± 0.04	96.55 ± 0.09	74.41 ± 0.71	98.55 ± 0.07	98.18 ± 0.09
		aAcc	99.24 ± 0.02	98.77 ± 0.06	**99.27 ± 0.02**	98.04 ± 0.07	98.79 ± 0.05	99.15 ± 0.05	97.41 ± 0.16	85.23 ± 0.41	98.81 ± 0.03	98.34 ± 0.08
	200	mIoU	97.99 ± 0.10	96.74 ± 0.24	98.15 ± 0.11	96.10 ± 0.26	98.10 ± 0.13	**98.46 ± 0.11**	94.39 ± 0.32	56.09 ± 0.69	96.96 ± 0.10	95.55 ± 0.09
		mAcc	99.38 ± 0.04	98.97 ± 0.08	99.40 ± 0.03	98.70 ± 0.06	99.39 ± 0.03	**99.51 ± 0.04**	98.04 ± 0.06	75.20 ± 0.63	98.97 ± 0.04	98.61 ± 0.04
		aAcc	99.49 ± 0.02	99.20 ± 0.06	99.52 ± 0.02	98.82 ± 0.07	99.47 ± 0.04	**99.57 ± 0.03**	98.54 ± 0.05	86.02 ± 0.33	99.21 ± 0.03	98.88 ± 0.03
Protocol B	25	mIoU	**35.42 ± 0.87**	24.45 ± 0.31	34.82 ± 1.67	34.83 ± 0.66	30.74 ± 0.25	30.40 ± 0.97	29.94 ± 0.77	32.28 ± 0.49	29.21 ± 0.80	14.85 ± 1.51
		mAcc	**49.17 ± 1.36**	37.50 ± 0.33	47.05 ± 1.15	47.32 ± 0.72	41.92 ± 0.41	43.42 ± 1.83	40.55 ± 0.81	46.26 ± 0.56	40.54 ± 0.89	27.39 ± 1.00
		aAcc	75.61 ± 1.21	51.98 ± 3.53	**76.53 ± 1.66**	75.85 ± 0.50	73.42 ± 0.54	71.67 ± 1.26	72.13 ± 0.90	72.25 ± 0.44	72.18 ± 0.91	39.32 ± 6.40
	50	mIoU	**35.54 ± 0.87**	23.65 ± 0.61	35.10 ± 2.63	34.80 ± 1.46	31.73 ± 0.33	30.99 ± 1.21	31.01 ± 1.15	33.85 ± 0.59	29.26 ± 0.77	15.32 ± 1.32
		mAcc	**48.70 ± 0.90**	35.90 ± 0.57	47.92 ± 2.10	47.12 ± 1.52	42.95 ± 0.54	43.85 ± 1.20	41.36 ± 1.27	47.65 ± 0.71	40.53 ± 0.94	28.08 ± 0.80
		aAcc	**76.67 ± 1.28**	50.22 ± 3.08	76.62 ± 1.82	75.78 ± 1.00	74.48 ± 0.31	72.34 ± 1.66	73.88 ± 1.00	74.07 ± 0.47	72.48 ± 0.60	40.59 ± 5.60
	100	mIoU	**35.54 ± 0.93**	23.86 ± 1.12	35.53 ± 2.47	34.95 ± 0.93	31.99 ± 0.62	30.56 ± 0.95	31.19 ± 0.96	33.97 ± 0.55	30.21 ± 0.70	15.68 ± 1.38
		mAcc	**48.58 ± 1.27**	36.66 ± 1.27	48.03 ± 2.16	46.85 ± 0.99	42.84 ± 0.81	43.18 ± 1.52	41.48 ± 0.98	47.43 ± 0.79	41.48 ± 0.81	28.40 ± 1.24
		aAcc	76.51 ± 1.86	50.69 ± 3.33	**76.97 ± 1.67**	76.28 ± 0.58	75.17 ± 0.44	72.24 ± 1.02	74.40 ± 0.79	74.54 ± 0.38	73.33 ± 0.55	41.06 ± 5.78
	200	mIoU	**35.59 ± 1.05**	25.39 ± 0.85	35.53 ± 2.86	35.06 ± 0.43	32.41 ± 0.60	30.83 ± 1.36	32.34 ± 0.96	34.58 ± 0.30	29.77 ± 0.56	15.58 ± 1.08
		mAcc	**49.44 ± 0.97**	37.69 ± 1.82	48.60 ± 2.46	46.79 ± 0.60	43.49 ± 1.01	43.70 ± 1.83	43.11 ± 1.33	47.91 ± 0.45	40.94 ± 0.72	28.56 ± 0.72
		aAcc	75.95 ± 1.84	52.57 ± 3.97	**77.08 ± 1.87**	76.50 ± 0.24	75.38 ± 0.27	72.11 ± 1.45	75.57 ± 1.01	75.14 ± 0.35	73.16 ± 0.37	40.84 ± 5.11
Protocol C	25	mIoU	31.28 ± 0.31	26.45 ± 0.80	**33.67 ± 0.84**	29.25 ± 1.17	27.81 ± 0.77	27.10 ± 1.65	25.49 ± 1.89	28.97 ± 0.46	22.08 ± 0.46	10.71 ± 0.80
		mAcc	43.04 ± 0.72	37.28 ± 1.24	**45.71 ± 1.26**	41.17 ± 1.72	38.74 ± 0.83	37.97 ± 1.91	36.57 ± 2.19	42.75 ± 0.45	32.03 ± 0.74	18.61 ± 1.51
		aAcc	72.49 ± 0.45	68.19 ± 0.49	**74.56 ± 0.66**	69.97 ± 0.93	69.49 ± 0.85	68.98 ± 1.72	65.59 ± 2.02	67.08 ± 0.65	64.77 ± 0.76	43.39 ± 3.07
	50	mIoU	31.34 ± 0.67	26.30 ± 1.59	**33.85 ± 0.88**	28.88 ± 0.73	29.62 ± 0.82	27.89 ± 1.50	25.34 ± 3.03	29.77 ± 0.46	22.05 ± 0.77	10.98 ± 0.70
		mAcc	42.73 ± 1.07	36.71 ± 1.93	**45.63 ± 1.23**	40.24 ± 1.11	40.62 ± 1.13	38.56 ± 1.77	36.30 ± 3.29	43.17 ± 0.26	31.77 ± 0.95	18.75 ± 1.66
		aAcc	72.93 ± 0.49	69.11 ± 1.13	**74.98 ± 0.43**	70.22 ± 0.54	71.78 ± 0.66	70.16 ± 1.45	66.02 ± 4.21	68.49 ± 0.93	65.29 ± 0.72	44.29 ± 2.33
	100	mIoU	32.13 ± 0.23	26.82 ± 0.94	**34.03 ± 1.46**	28.91 ± 0.96	29.81 ± 0.43	27.64 ± 1.12	27.15 ± 1.06	30.57 ± 0.80	21.96 ± 0.44	11.19 ± 0.51
		mAcc	43.43 ± 0.63	37.59 ± 0.93	**45.80 ± 1.84**	39.82 ± 1.33	40.58 ± 0.40	38.21 ± 1.51	38.18 ± 1.23	43.77 ± 0.89	31.56 ± 0.52	18.90 ± 1.13
		aAcc	73.56 ± 0.49	69.52 ± 0.90	**75.45 ± 0.80**	70.63 ± 0.82	72.30 ± 0.40	70.42 ± 0.98	68.65 ± 0.99	69.75 ± 1.09	65.52 ± 0.68	45.52 ± 2.88
	200	mIoU	32.28 ± 0.59	26.72 ± 2.01	**34.54 ± 0.71**	28.77 ± 0.85	30.15 ± 0.54	27.43 ± 1.20	27.15 ± 0.99	30.91 ± 0.34	21.64 ± 1.01	11.34 ± 0.24
		mAcc	43.95 ± 1.08	37.20 ± 2.77	**46.47 ± 0.99**	39.47 ± 1.18	40.85 ± 0.87	37.70 ± 1.31	37.82 ± 1.15	43.78 ± 0.27	31.01 ± 1.26	19.28 ± 1.07
		aAcc	73.71 ± 0.63	69.60 ± 1.61	**75.50 ± 0.36**	70.68 ± 0.66	72.69 ± 0.21	70.38 ± 1.42	68.97 ± 1.07	70.36 ± 0.52	65.38 ± 0.82	46.44 ± 1.83

to Protocol B, increasing the number of training samples results in only limited performance gains for most methods. Moreover, the relative performance of different architectures changes considerably compared with Protocol A; for example, several methods achieving high accuracy under random sampling show substantially lower performance under Protocol C. These results demonstrate that hyperspectral classification performance varies across different evaluation protocols. High accuracy under conventional random sampling does not translate into comparable performance under spatially disjoint evaluation settings.

Statistical significance analysis. At $N = 25$, we compared protocol-level mIoU using two-sided paired t-tests, matching runs by their seed identifiers (42–46). Holm correction was applied jointly across the 30 comparisons (ten methods and three protocol pairs). We report the mean paired difference, 95% confidence interval, and paired effect size d_z. At a significance level of 0.05, 29 of the 30 comparisons remained significant after correction; the non-significant comparison was B versus C for SegFormer. These tests describe variation across five runs under the specified partitions and do not quantify uncertainty across independent scenes. Detailed results are provided in Table S3 of the Supplementary Material.

2) Qualitative Analysis: Figs. 5–7 present representative classification maps of different methods under the three protocols at $N = 25$. Under Protocol A, most methods produce predictions that closely match the ground truth, with major crop parcels correctly identified and their spatial structures largely preserved. The remaining errors are mainly observed in small parcels and local boundary regions.

When spatially disjoint evaluation is introduced in Protocols B and C, noticeably larger discrepancies appear in the classification maps. Under Protocol B, errors are particularly evident in the right-half testing regions, where several homogeneous crop parcels are partially or largely assigned to incorrect

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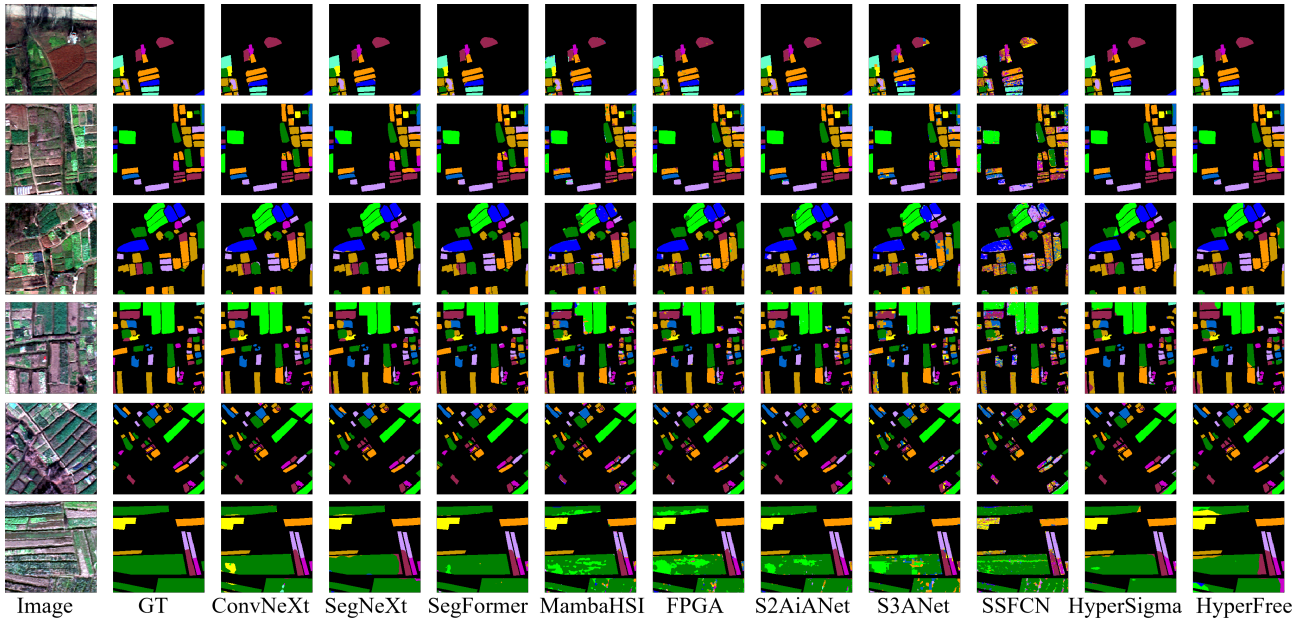


Fig. 5. Visualization of classification results of different models on WHU-Hi+ under Protocol A.

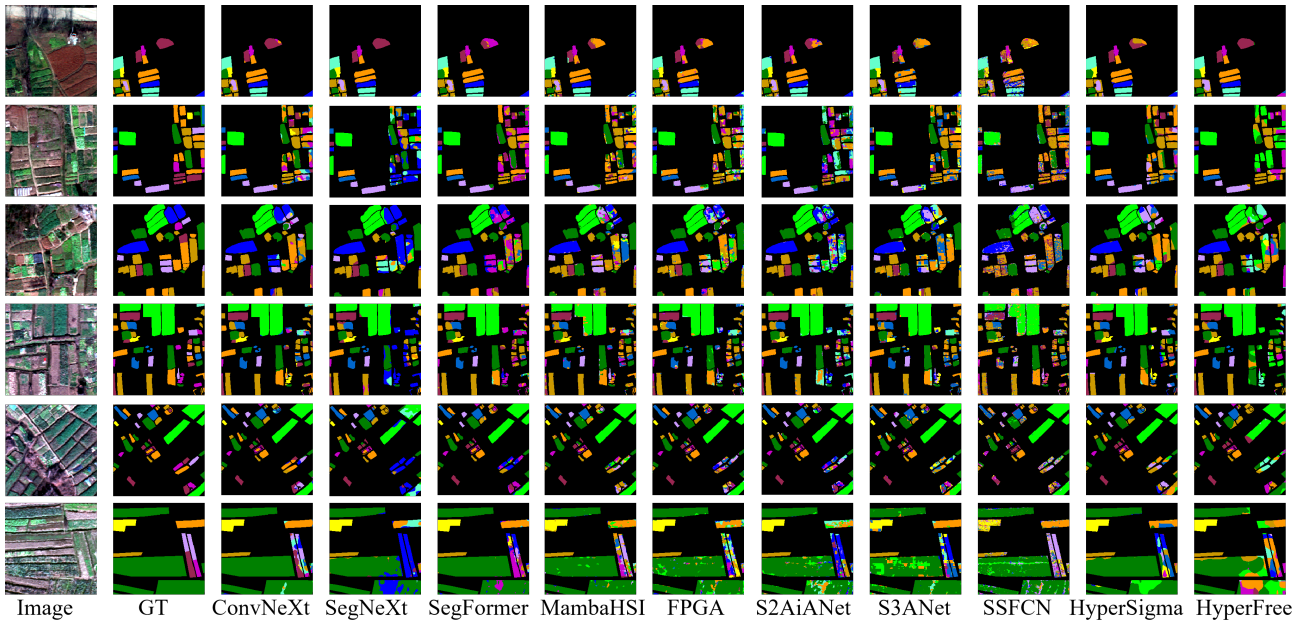


Fig. 6. Visualization of classification results of different models on WHU-Hi+ under Protocol B.

classes. Under Protocol C, parcel-level misclassifications are also frequently observed, and some methods produce fragmented class assignments within otherwise continuous crop regions. These examples illustrate errors in class assignment within the displayed labeled regions; they do not independently establish parcel-boundary accuracy.

A direct comparison of the three protocols on the same representative image tile is further provided in Fig. 8. The cross-protocol comparison shows that predictions under Protocol A are generally more consistent with the ground truth, whereas Protocols B and C exhibit more frequent parcel-level misclassifications and within-parcel inconsistencies. These differences are particularly evident in several large homogeneous crop regions, where methods that produce relatively consistent predictions under Protocol A show substantial label changes under the spatially separated protocols.

Together, these qualitative results further illustrate the sensitivity of classification outcomes to the adopted train–test spatial partitioning strategy and complement the quantitative comparisons in Table V.

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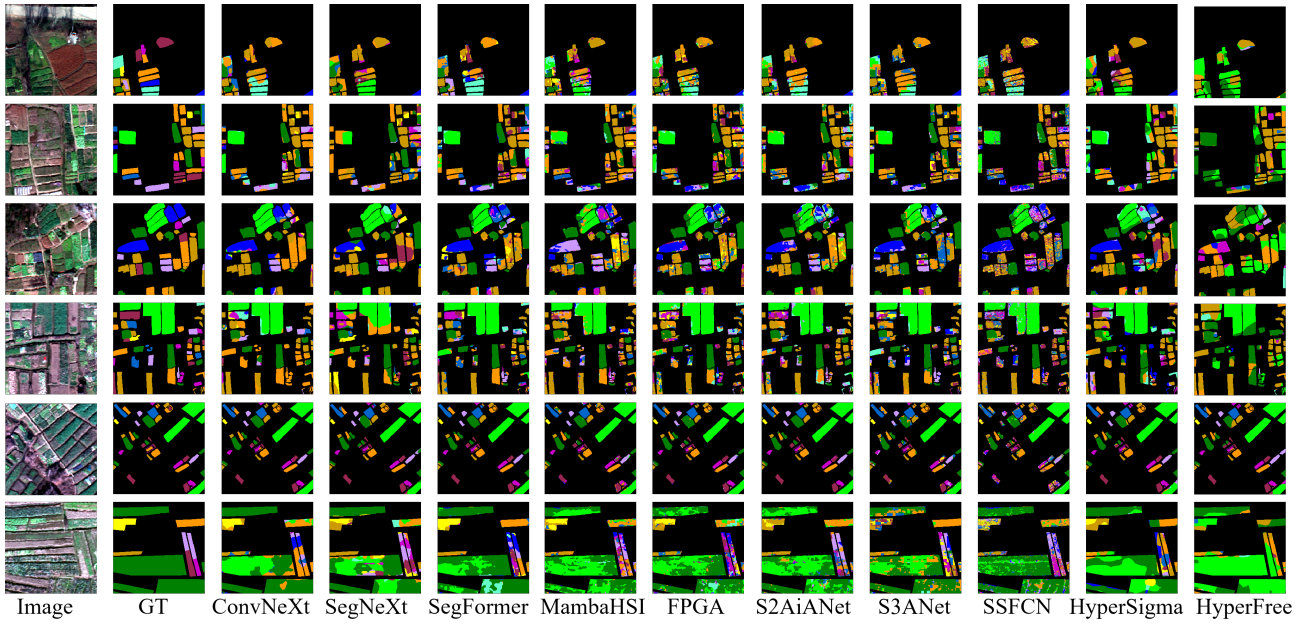


Fig. 7. Visualization of classification results of different models on WHU-Hi+ under Protocol C.

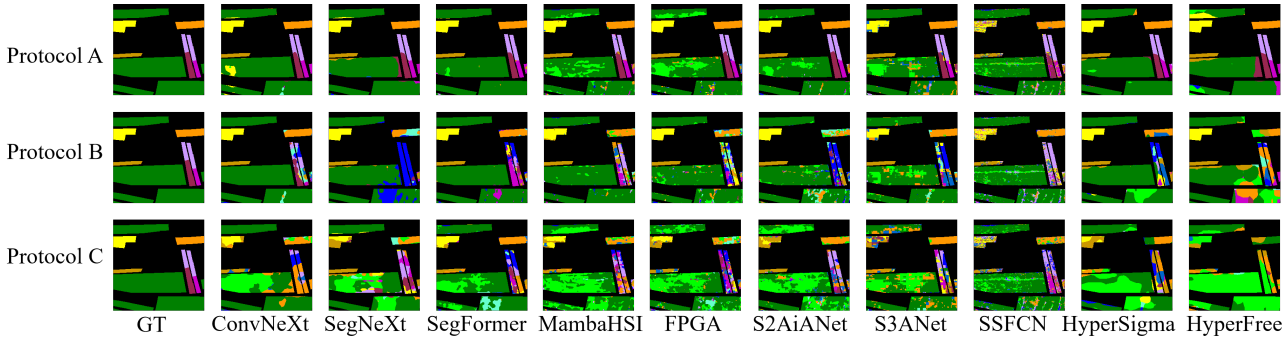


Fig. 8. Qualitative comparison of classification results for the same representative image tile under Protocols A–C.

V. Discussion

A. Characteristics and Challenges of WHU-Hi+

WHU-Hi+ combines several characteristics that make fine-grained hyperspectral crop classification challenging. Its 105 image tiles, derived from multi-strip UAV observations, provide spatially distributed samples for evaluating different training–test partitioning protocols. The 11 crop categories exhibit considerable inter-class spectral similarity and intra-class spectral variability, increasing the difficulty of distinguishing closely related crops across spatial locations. In addition, the pronounced class imbalance, with rape accounting for more than half of the labeled pixels, introduces substantial differences in sample availability among categories. Together, these characteristics provide a challenging benchmark for examining classification performance under different spatial partitioning strategies.

1) Impact of Train–Test Spatial Partitioning: The results across Protocols A–C show substantial performance differences between evaluation settings. Most methods achieve high accuracy under Protocol A and lower accuracy under Protocols B and C. These protocols differ in spatial arrangement, the number and distribution of available training labels, and test-set composition. Together, they provide complementary evaluation settings for assessing model performance under varying spatial and training conditions.

The observed performance differences highlight the importance of spatial partitioning in hyperspectral benchmark evaluation. High accuracy under random pixel sampling does not necessarily indicate comparable performance under spatially disjoint settings. Evaluating models across complementary partitioning protocols therefore provides a more comprehensive characterization of their classification performance and robustness.

2) Category-Level Performance: The impact of spatial partitioning varies across crop categories, as shown in Table VI. Complementary per-class accuracy results are provided in Table S1 of the Supplementary Material. Under Protocol A, most classes achieve relatively high IoU, particularly for the stronger-performing models, whereas the transition to Protocols B and C produces markedly different degrees of degradation among classes. This category-dependent response indicates that the overall performance gap across protocols is

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TABLE VI
PER-CLASS IOU (%) OF TEN MODELS ON WHU-HI+ UNDER THREE PROTOCOLS WITH $N = 25$. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE SEEDS.

Protocol	Crop class	ConvNeXt	SegNeXt	SegFormer	MambaHSI	FPGA	S^2AiANet	S^3ANet	SSFCN	HyperSIGMA	HyperFree
Protocol A	Rape	99.07 ± 0.28	98.06 ± 0.50	**99.09 ± 0.27**	95.76 ± 0.26	96.51 ± 0.30	97.75 ± 0.31	94.49 ± 0.99	88.29 ± 0.60	98.36 ± 0.39	94.79 ± 1.39
	Pakchoi	**87.48 ± 1.34**	84.54 ± 1.52	87.11 ± 0.97	74.77 ± 1.04	74.91 ± 1.69	81.96 ± 1.29	67.59 ± 3.11	33.11 ± 0.79	83.10 ± 2.32	77.85 ± 2.23
	Chinese cabbage	90.48 ± 0.71	88.77 ± 1.14	**90.58 ± 0.59**	80.29 ± 0.36	81.52 ± 1.09	87.47 ± 0.39	74.29 ± 1.61	55.19 ± 1.21	84.63 ± 1.15	78.18 ± 1.92
	Greens	89.50 ± 0.20	87.73 ± 0.59	**89.72 ± 0.86**	78.62 ± 0.76	77.18 ± 1.20	85.17 ± 0.64	64.72 ± 1.96	46.84 ± 1.09	85.24 ± 0.85	79.88 ± 2.39
	Nori moss	94.63 ± 0.45	92.04 ± 0.64	**95.06 ± 0.43**	89.34 ± 1.38	89.52 ± 1.02	92.44 ± 0.55	81.50 ± 5.89	76.87 ± 1.27	89.28 ± 0.48	81.97 ± 1.04
	Lettuce	88.12 ± 0.98	85.63 ± 2.32	**88.92 ± 0.92**	78.63 ± 1.26	81.14 ± 0.54	86.33 ± 1.33	80.34 ± 0.98	38.03 ± 0.97	85.24 ± 1.15	79.69 ± 1.97
	Broad bean	**95.51 ± 0.83**	89.87 ± 2.81	94.79 ± 1.00	81.22 ± 0.97	80.81 ± 0.21	89.89 ± 0.67	74.84 ± 2.11	47.30 ± 2.22	93.56 ± 0.81	91.42 ± 1.23
	Pea	**94.62 ± 0.87**	92.06 ± 1.73	93.51 ± 1.73	83.69 ± 1.47	81.32 ± 1.79	90.14 ± 1.33	78.29 ± 1.61	43.88 ± 1.21	91.91 ± 0.98	90.95 ± 1.84
	Turnip	89.68 ± 0.41	87.32 ± 1.18	**91.23 ± 0.53**	77.73 ± 0.49	74.62 ± 2.05	85.15 ± 1.16	44.88 ± 3.50	36.00 ± 1.40	87.35 ± 1.06	84.60 ± 2.23
	Mustard	93.07 ± 1.45	91.86 ± 1.46	**94.00 ± 0.94**	83.87 ± 0.84	79.94 ± 3.47	90.47 ± 0.93	51.69 ± 2.95	35.62 ± 2.65	93.54 ± 0.45	90.14 ± 1.00
	Cauliflower	**87.88 ± 0.83**	84.79 ± 2.77	87.11 ± 1.83	75.79 ± 0.70	79.89 ± 2.03	84.41 ± 2.18	69.64 ± 3.75	33.36 ± 0.80	82.91 ± 2.22	78.09 ± 3.42
Protocol B	Rape	90.12 ± 1.76	55.95 ± 6.17	**91.65 ± 0.77**	89.26 ± 0.58	90.80 ± 0.54	87.98 ± 2.31	90.73 ± 1.07	85.78 ± 0.68	88.35 ± 1.01	40.20 ± 10.53
	Pakchoi	**18.96 ± 1.23**	16.06 ± 2.78	17.54 ± 1.90	14.80 ± 1.15	12.94 ± 1.51	14.91 ± 1.36	13.87 ± 2.51	14.55 ± 0.82	12.44 ± 1.20	11.31 ± 0.55
	Chinese cabbage	**54.11 ± 1.92**	42.01 ± 3.18	50.52 ± 2.49	45.74 ± 1.48	42.00 ± 2.80	42.72 ± 2.02	45.09 ± 2.39	46.38 ± 0.95	39.65 ± 2.82	21.45 ± 3.00
	Greens	28.30 ± 2.58	18.05 ± 2.27	31.34 ± 2.18	**32.26 ± 1.19**	22.85 ± 2.11	22.88 ± 3.70	23.25 ± 1.44	31.07 ± 0.60	24.55 ± 1.69	11.58 ± 1.05
	Nori moss	77.92 ± 1.83	66.07 ± 3.11	78.55 ± 3.49	**79.50 ± 1.92**	77.79 ± 0.94	72.82 ± 1.58	74.02 ± 5.46	73.30 ± 1.22	64.35 ± 2.08	21.35 ± 0.83
	Lettuce	**22.81 ± 2.00**	13.60 ± 1.19	18.28 ± 4.05	22.22 ± 2.80	15.88 ± 1.18	17.18 ± 3.46	17.77 ± 1.50	18.91 ± 1.13	14.97 ± 2.00	8.63 ± 1.72
	Broad bean	25.34 ± 2.94	11.21 ± 1.57	**30.17 ± 8.67**	26.96 ± 1.35	26.75 ± 2.26	25.94 ± 1.02	24.62 ± 4.07	24.04 ± 2.14	23.55 ± 4.45	9.99 ± 1.23
	Pea	28.62 ± 5.27	17.19 ± 4.71	**29.48 ± 1.06**	26.85 ± 1.27	21.32 ± 1.39	22.12 ± 2.57	20.12 ± 2.47	22.65 ± 0.69	25.33 ± 2.89	22.65 ± 0.79
	Turnip	**14.45 ± 3.84**	8.96 ± 3.18	13.20 ± 5.04	11.97 ± 1.83	8.15 ± 0.99	8.61 ± 1.76	8.31 ± 0.86	12.42 ± 1.56	5.91 ± 1.52	3.60 ± 0.82
	Mustard	10.24 ± 1.24	10.93 ± 3.83	9.61 ± 1.71	**13.61 ± 2.37**	6.60 ± 0.61	6.58 ± 1.16	3.10 ± 1.10	8.65 ± 2.39	7.29 ± 2.57	6.88 ± 1.62
	Cauliflower	18.75 ± 3.44	8.90 ± 1.12	12.68 ± 3.61	**19.94 ± 1.94**	13.11 ± 1.37	12.66 ± 1.42	8.50 ± 1.43	17.36 ± 1.14	14.88 ± 1.81	5.76 ± 0.51
Protocol C	Rape	89.93 ± 0.87	85.87 ± 0.98	**92.00 ± 1.33**	85.99 ± 1.19	87.55 ± 0.92	86.47 ± 1.87	86.01 ± 1.58	81.27 ± 1.08	84.03 ± 1.44	52.90 ± 5.42
	Pakchoi	13.28 ± 2.28	6.29 ± 1.97	**14.58 ± 1.94**	11.61 ± 0.88	9.21 ± 1.24	9.88 ± 1.40	8.62 ± 1.63	11.11 ± 0.47	6.56 ± 0.83	1.71 ± 1.08
	Chinese cabbage	**49.52 ± 1.21**	41.93 ± 1.88	49.22 ± 1.88	41.33 ± 0.68	42.09 ± 0.78	41.35 ± 2.11	39.02 ± 2.90	42.06 ± 1.29	35.06 ± 0.63	19.89 ± 1.52
	Greens	23.70 ± 1.58	17.90 ± 2.52	**31.37 ± 1.87**	26.49 ± 1.04	19.28 ± 1.33	19.25 ± 1.57	17.61 ± 1.88	26.54 ± 0.54	17.02 ± 1.34	8.59 ± 2.81
	Nori moss	74.76 ± 2.99	65.15 ± 1.79	**76.14 ± 2.56**	69.29 ± 3.85	66.82 ± 3.12	65.16 ± 5.07	51.44 ± 6.56	66.50 ± 1.88	46.36 ± 3.22	14.97 ± 2.59
	Lettuce	15.74 ± 2.62	9.08 ± 3.29	**16.28 ± 2.98**	10.09 ± 1.72	8.93 ± 2.10	11.36 ± 2.20	9.54 ± 2.12	11.24 ± 0.87	9.48 ± 1.01	4.14 ± 2.40
	Broad bean	18.22 ± 4.05	16.41 ± 3.45	23.96 ± 4.74	16.27 ± 1.70	**25.72 ± 2.31**	24.47 ± 3.57	21.86 ± 2.90	21.48 ± 0.77	12.65 ± 1.80	3.24 ± 2.82
	Pea	**26.80 ± 2.95**	21.06 ± 3.24	26.58 ± 2.19	22.00 ± 2.39	23.84 ± 1.06	21.55 ± 2.08	22.71 ± 6.32	22.33 ± 1.51	19.85 ± 1.77	6.25 ± 2.01
	Turnip	8.93 ± 1.90	4.68 ± 2.08	13.33 ± 3.74	13.49 ± 1.62	7.87 ± 1.78	5.72 ± 1.95	6.42 ± 2.07	**13.57 ± 1.08**	2.49 ± 1.67	2.63 ± 2.02
	Mustard	6.03 ± 3.07	**11.10 ± 5.91**	10.06 ± 1.95	9.56 ± 1.19	4.48 ± 2.11	3.18 ± 1.20	3.29 ± 1.38	7.38 ± 1.31	3.72 ± 2.23	0.56 ± 0.92
	Cauliflower	**17.20 ± 1.56**	11.51 ± 2.73	16.79 ± 4.45	15.60 ± 3.22	10.11 ± 0.74	9.65 ± 4.69	13.87 ± 1.59	15.26 ± 1.28	5.71 ± 1.08	2.89 ± 1.03

not uniformly distributed over the 11 crop categories.

Rape and nori moss remain relatively well recognized under spatially disjoint evaluation. At $N = 25$, for example, rape retains IoU values above 80% for most models under Protocols B and C, while nori moss also consistently achieves higher IoU than most other crop categories. Their relatively stable performance contrasts with the substantial degradation observed for many other classes, showing that spatial separation does not affect all categories to the same extent.

Several fine-grained crop categories become particularly difficult under Protocols B and C. Pakchoi, turnip, and mustard exhibit pronounced IoU reductions across most models, while Chinese cabbage generally retains higher performance despite also decreasing from Protocol A. These results are consistent with the combined challenges of inter-class spectral similarity, intra-class variability, and uneven class distributions identified in WHU-Hi+. However, the different behavior among individual categories suggests that their performance cannot be attributed to any single factor alone.

3) Model Behavior under Different Protocols: The evaluated models exhibit distinct performance patterns across spatial partitioning protocols. ConvNeXt and SegFormer consistently rank among the leading methods, whereas several other models show substantial changes in relative performance. For example, at $N = 25$, SSFCN achieves an mIoU of 48.59% under Protocol A but retains 32.28% and 28.97% under Protocols B and C, respectively. In contrast, HyperFree achieves 84.32% under Protocol A but decreases to 14.85% and 10.71% under Protocols B and C. These contrasting results show that the relative performance of the evaluated implementations changes across protocols. They do not establish an architecture-wide ranking independent of initialization, decoder design, and optimization settings.

The observed differences raise the question of how hyperspectral classification models can maintain reliable discrimination across spatially separated labeled regions. In WHU-Hi+, fine-grained crop categories are spectrally similar and exhibit intra-class spectral variability. Consequently, learning discriminative spectral–spatial representations that generalize across spatially disjoint training and testing regions is challenging. These findings highlight the need to improve the robustness of spectral–spatial feature learning to spatial variations in hyperspectral data.

B. Limitations and Future Extensions

The current WHU-Hi+ benchmark is limited to a single study area and a single acquisition date. Although its multi-strip UAV observations provide broad spatial coverage within the study area, the dataset does not include geographically distinct study regions or repeated observations that systematically sample temporal changes in crop growth. The current protocols therefore focus on the effects of training–test spatial partitioning within the collected scene rather than cross-region or multi-temporal evaluation.

The crop-focused taxonomy also limits the scope of the current benchmark. WHU-Hi+ is designed for fine-grained agricultural hyperspectral classification and contains 11 crop categories, without heterogeneous land-cover classes such as buildings, roads, and other artificial surfaces. Consequently, the benchmark is well suited to evaluating fine-grained crop discrimination under different spatial partitioning strategies, but does not represent the full diversity of land-cover types encountered in more general hyperspectral remote-sensing applications.

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Future extensions will broaden both the spatial–temporal coverage and the category diversity of WHU-Hi+. Additional observations from different regions and acquisition periods would enable evaluation across geographic and phenological variations, while incorporating more heterogeneous agricultural scenes and land-cover categories would extend the benchmark to a wider range of remote-sensing conditions. These extensions would complement the current benchmark while preserving its focus on large-scale, fine-grained hyperspectral analysis.

VI. Conclusion

This study presents WHU-Hi+, a large-scale UAV-based H^2 hyperspectral benchmark for fine-grained image classification, comprising 105 non-overlapping image tiles and 11 crop categories with pronounced class imbalance, inter-class spectral similarity, intra-class spectral variability, and heterogeneous spatial distributions. Three evaluation protocols are established to characterize model performance under different train–test spatial partitioning settings. Benchmark results from ten representative deep learning models show that most methods achieve high accuracy under conventional random sampling but exhibit substantial performance degradation under spatially disjoint protocols when spatial separation is introduced, with relative model performance also varying across protocols. These results show that classification performance depends on the evaluation protocol, which jointly determines spatial partitioning, training-sample availability, and test-set composition. WHU-Hi+ therefore provides a challenging benchmark for evaluating fine-grained hyperspectral image classification under diverse spatial sampling conditions and for studying limitations related to spectral confusion, class imbalance, and spatially disjoint evaluation. Future work will extend the dataset to more diverse regions, acquisition periods, and land-cover categories.

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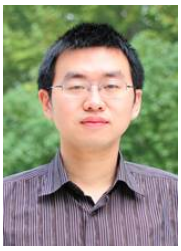


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